DSC 206 — Homework 1 Solutions

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Problem 1 (5 pts)

Claim: For the rank-k approximation

$$ A_k = \sum_{i=1}^k \sigma_i u_i v_i^T, $$

we have

$$ |(A - A_k)x|2 |le |sigma{k+1}|x|_2. $$

Proof.

$A - A_k = \sum_{i=k+1}^r \sigma_i u_i v_i^T$, where $r = \operatorname{rank}(A)$.

Expand x in the orthonormal basis ${v_1, \ldots, v_n}$:

$$ x = \sum_j (v_j^T x), v_j. $$

Then $$ (A - A_k)x = \sum_{i=k+1}^r \sigma_i u_i v_i^T \Bigl(\sum_j (v_j^T x) v_j\Bigr) = \sum_{i=k+1}^r \sigma_i (v_i^T x) u_i. $$

Since ${u_i}$ are orthonormal, $$ |(A-A_k)x|2^2 = |sum{i=k+1}^r \sigma_i^2 (v_i^T x)^2 \le \sigma_{k+1}^2 \sum_{i=k+1}^r (v_i^T x)^2 \le \sigma_{k+1}^2 \sum_j (v_j^T x)^2 = \sigma_{k+1}^2 |x|_2^2. $$

The first inequality uses $\sigma_i \le \sigma_{k+1}$ for $i \ge k+1$; the second uses Parseval's identity. Taking square roots gives the result. $\square$

Key insight: The residual $A - A_k$ only acts on the tail singular directions ${v_{k+1}, \ldots}$, and its operator norm equals σ_{k+1}, the largest discarded singular value.

Problem 2 (15 pts)

$$ A = \begin{pmatrix} 1&2\ -1&2\ 1&-2\ -1&-2 \end{pmatrix} $$

(a) Power method, $k=3$ steps from $x^{(0)}=[1,2]^T$

First compute $B = A^T A$:

$$ A^T A = \begin{pmatrix}1&-1&1&-1\2&2&-2&-2\end{pmatrix} \begin{pmatrix}1&2\-1&2\1&-2\-1&-2\end{pmatrix} = \begin{pmatrix}4&0\0&16\end{pmatrix}. $$

Power iteration: $x^{(t+1)} \propto Bx^{(t)}$.

Start from

$$ x^{(0)} = \begin{pmatrix}1\2\end{pmatrix}. $$

Step 1:

$$ Bx^{(0)} = \begin{pmatrix}4\32\end{pmatrix}, \qquad x^{(1)} = \frac{1}{\sqrt{65}}\begin{pmatrix}1\8\end{pmatrix}. $$

Step 2:

$$ Bx^{(1)} = \frac{1}{\sqrt{65}}\begin{pmatrix}4\128\end{pmatrix}, \qquad x^{(2)} = \frac{1}{\sqrt{1025}}\begin{pmatrix}1\32\end{pmatrix}. $$

Step 3:

$$ Bx^{(2)} = \frac{1}{\sqrt{1025}}\begin{pmatrix}4\512\end{pmatrix}, \qquad x^{(3)} = \frac{1}{\sqrt{16385}}\begin{pmatrix}1\128\end{pmatrix}. $$

After 3 steps: $x^{(3)} = [1,128]^T/\sqrt{16385}$.

Convergence note: The ratio of components $128/1 = 128 = 2^7$ is growing toward ∞, converging to $v_1 = [0,1]^T$ (see part b).

(b) Actual SVD of A

Since $A^TA = \operatorname{diag}(4,16)$, the eigensystem is immediate.

For $i=1$,

$$ \sigma_1 = \sqrt{16} = 4, \qquad v_1 = \begin{pmatrix}0\1\end{pmatrix}, \qquad u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{4}\begin{pmatrix}2\2\-2\-2\end{pmatrix} = \begin{pmatrix}\tfrac12\\\tfrac12\-\tfrac12\-\tfrac12\end{pmatrix}. $$

For $i=2$,

$$ \sigma_2 = \sqrt{4} = 2, \qquad v_2 = \begin{pmatrix}1\0\end{pmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{2}\begin{pmatrix}1\-1\1\-1\end{pmatrix} = \begin{pmatrix}\tfrac12\-\tfrac12\\\tfrac12\-\tfrac12\end{pmatrix}. $$

(c) Rotation

Let

$$ R_\theta = \begin{pmatrix} \cos\theta&\sin\theta\ -\sin\theta&\cos\theta \end{pmatrix} $$

(clockwise rotation).

c.1 The problem defines $\hat A^T = R_\theta A^T$, so $\hat A = A R_\theta^T$.

$$ \hat A = A\begin{pmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{pmatrix} = \begin{pmatrix} \cos\theta+2\sin\theta & -\sin\theta+2\cos\theta\ -\cos\theta+2\sin\theta & \sin\theta+2\cos\theta\ \cos\theta-2\sin\theta & -\sin\theta-2\cos\theta\ -\cos\theta-2\sin\theta & \sin\theta-2\cos\theta \end{pmatrix}. $$

c.2 Write $A = U\Sigma V^T$; then $\hat A = U\Sigma(R_\theta V)^T$.

Thus **the singular values are unchanged**: $\hat\sigma_1 = 4$, $\hat\sigma_2 = 2$.

The **new first right singular vector** is

$$ \hat v_1 = R_\theta v_1 = R_\theta\begin{pmatrix}0\1\end{pmatrix} = \begin{pmatrix}\sin\theta\\\cos\theta\end{pmatrix}. $$

In general, the singular values are rotation-invariant, and the right singular vectors rotate with the data.

Problem 3 (10 pts)

Let A be $m\times n$ with unit-norm rows ${a_i}$.

Part 1 — single synthetic document

We want y (unit norm) maximizing

$$ \sum_i (a_i^T y)^2 = |Ay|_2^2 = y^T A^T A, y. $$

By the Rayleigh quotient, the maximum is the **largest eigenvalue** of A^TA, achieved at its leading eigenvector, which is the **first right singular vector v_1** of A.

This v_1 is the principal direction of the document cloud; it represents the topic most coherently expressed across all documents.

Part 2 — k orthogonal synthetic documents

Maximize

$$ \sum_{j=1}^k |Ay_j|*2^2 = |sum*{j=1}^k y_j^T A^T A, y_j $$

subject to ${y_j}$ orthonormal.

By the variational characterization of eigenvalues (Fan trace maximization),

$$ \max_{{y_j}\text{ o.n.}} \sum_{j=1}^k y_j^T(A^TA)y_j = \lambda_1+\lambda_2+\cdots+\lambda_k, $$

achieved at the **first k right singular vectors** ${v_1, \ldots, v_k}$.

These are also the first k columns of V in the SVD $A = U\Sigma V^T$.

Problem 4 (5 pts)

$$ B = \begin{pmatrix}2&1\1&3\end{pmatrix}, \quad x^{(0)} = [1,1]^T. $$

$B^T B = B^2$ (since B is symmetric):

$$ B^2 = \begin{pmatrix}5&5\5&10\end{pmatrix}. $$

Three power iterations

Starting from

$$ x^{(0)} = \begin{pmatrix}1\1\end{pmatrix}, $$

the first three iterations are:

$$ B^2x^{(0)} = \begin{pmatrix}10\15\end{pmatrix}, \qquad x^{(1)} = \frac{1}{\sqrt{13}}\begin{pmatrix}2\3\end{pmatrix}, $$

$$ B^2x^{(1)} = \frac{1}{\sqrt{13}}\begin{pmatrix}25\40\end{pmatrix} = \frac{5}{\sqrt{13}}\begin{pmatrix}5\8\end{pmatrix}, \qquad x^{(2)} = \frac{1}{\sqrt{89}}\begin{pmatrix}5\8\end{pmatrix}, $$

$$ B^2x^{(2)} = \frac{5}{\sqrt{89}}\begin{pmatrix}13\21\end{pmatrix}, \qquad x^{(3)} = \frac{1}{\sqrt{610}}\begin{pmatrix}13\21\end{pmatrix}. $$

First right singular vector estimate: $v_1 \approx [13,21]^T/\sqrt{610}$.

(Ratio $21/13 \approx 1.615 \approx \varphi = (1+\sqrt5)/2$, converging rapidly.)

First singular value

$$ \sigma_1 \approx |Bx^{(3)}|_2 = \left|B\frac{[13,21]^T}{\sqrt{610}}\right|_2 = \frac{|[47,76]^T|_2}{\sqrt{610}} = \sqrt{\frac{7985}{610}} \approx 3.618. $$

Exact: eigenvalues of B are $\lambda = \frac{5\pm\sqrt5}{2}$, so $\sigma_1 = \frac{5+\sqrt5}{2}$.

First left singular vector

$$ u_1 = \frac{Bv_1}{\sigma_1} \approx \frac{[47,76]^T}{\sqrt{7985}}.$$

Exact: $u_1 = v_1$ (since B is symmetric positive definite, $U=V$ in its SVD).

Second singular vector

$v_2 \perp v_1$ and is the other eigenvector:

$$ v_2 = \frac{[-(1+\sqrt5),,2]^T}{\sqrt{10+2\sqrt5}}, \qquad \sigma_2 = \frac{5-\sqrt5}{2} \approx 1.382. $$

Final SVD

$$ B = U\Sigma V^T, \quad U = V = \frac{1}{\sqrt{10+2\sqrt5}} \begin{pmatrix} 2 & -(1+\sqrt5)\ 1+\sqrt5 & 2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix}\frac{5+\sqrt5}{2}&0\0&\frac{5-\sqrt5}{2}\end{pmatrix}. $$

Problem 5 (12 pts)

$$ M = \begin{pmatrix} 1&1\ 2&4\ 2&4\ 3&9 \end{pmatrix}. $$

(a) M^TM and MM^T

$$ M^TM = \begin{pmatrix} 1^2+2^2+2^2+3^2 & 1\cdot1+2\cdot4+2\cdot4+3\cdot9\ \cdot & 1^2+4^2+4^2+9^2 \end{pmatrix} = \begin{pmatrix}18&44\44&114\end{pmatrix}. $$

MM^T is 4×4 with (i,j) entry equal to $m_i \cdot m_j$:

$$ MM^T = \begin{pmatrix} 2&6&6&12\ 6&20&20&42\ 6&20&20&42\ 12&42&42&90 \end{pmatrix}. $$

(Rows 2 and 3 of M are identical, so rows and columns 2 and 3 of MM^T are identical.)

(b) Eigenpairs of M^TM

$$ \det(M^TM - \lambda I) = \lambda^2 - 132\lambda + 116 = 0. $$

$$ \lambda = 66 \pm 4\sqrt{265}. $$

For $\lambda_1 = 66+4\sqrt{265}$, solve $(18-\lambda_1)v_1 + 44v_2 = 0$, giving

$$ v_1 \propto \begin{pmatrix}11\12+\sqrt{265}\end{pmatrix}, \quad \text{normalized by } \sqrt{530+24\sqrt{265}}. $$

For $\lambda_2 = 66-4\sqrt{265}$,

$$ v_2 \propto \begin{pmatrix}11\12-\sqrt{265}\end{pmatrix}, \quad \text{normalized by } \sqrt{530-24\sqrt{265}}. $$

(Note $|sqrt{265}|approx 16.28$, so $|lambda_1|approx 131.1$, $|lambda_2|approx 0.88$, and v_2 has a negative second component.)

(c) Eigenvalues of MM^T

MM^T (a 4×4 matrix of rank at most 2) shares its **nonzero** eigenvalues with M^TM. Therefore,

$$ \text{eigenvalues of } MM^T = {66+4\sqrt{265},; 66-4\sqrt{265},; 0,; 0}. $$

Problem 6 (6 pts)

Network (a)

Graph: 4-node chain with wrap-around (reading the edge labels in Figure 1):

- Node 1: $P(1\to1)=0.8$, $P(1\to2)=0.2$
- Node 2: $P(2\to1)=0.6$, $P(2\to3)=0.4$
- Node 3: $P(3\to2)=0.5$, $P(3\to4)=0.5$
- Node 4: $P(4\to3)=0.5$, $P(4\to1)=0.5$

Solve $\pi P = \pi$:

From $\pi_4 = 0.5\pi_3$ and $0.75\pi_3 = 0.4\pi_2$ and $(11/15)\pi_2 = (1/5)\pi_1$:

$$\pi_2 = \frac{3}{11}\pi_1, \quad \pi_3 = \frac{8}{55}\pi_1, \quad \pi_4 = \frac{4}{55}\pi_1.$$

Normalizing ($\pi_1\cdot\frac{82}{55}=1$):

$$\boxed{\pi_1 = \frac{55}{82},\quad \pi_2 = \frac{15}{82},\quad \pi_3 = \frac{8}{82},\quad \pi_4 = \frac{4}{82}.}$$

Network (b)

Graph: 4-node diamond (T=top, L=left, R=right, B=bottom):

- T: $P(T\to T)=0.1$, $P(T\to L)=0.3$, $P(T\to R)=0.6$
- L: $P(L\to T)=0.5$, $P(L\to L)=0.2$, $P(L\to B)=0.3$
- R: $P(R\to T)=0.5$, $P(R\to R)=0.2$, $P(R\to B)=0.3$
- B: $P(B\to L)=0.6$, $P(B\to R)=0.3$, $P(B\to B)=0.1$

Solve $\pi P = \pi$:

From equations: $$0.9\pi_T = 0.5(\pi_L+\pi_R), \quad 0.9\pi_B = 0.3(\pi_L+\pi_R)$$ so $\pi_B = \frac{1}{3}\cdot\frac{0.9}{0.5}\cdot\pi_T \cdot\frac{1}{2}$... more directly: $\pi_B = 0.6\pi_T$.

Then $0.8\pi_L = 0.66\pi_T$ giving $\pi_L = \frac{33}{40}\pi_T$ and $0.8\pi_R = 0.78\pi_T$ giving $\pi_R = \frac{39}{40}\pi_T$.

Normalizing ($\pi_T\cdot\frac{17}{5}=1$... wait: $1+\frac{33}{40}+\frac{39}{40}+\frac{3}{5} = \frac{40+33+39+24}{40}=\frac{136}{40}=\frac{17}{5}$):

$$\boxed{\pi_T = \frac{5}{17} = \frac{40}{136},\quad \pi_L = \frac{33}{136},\quad \pi_R = \frac{39}{136},\quad \pi_B = \frac{3}{17} = \frac{24}{136}.}$$

Problem 7 (4 pts)

Example: 2-state Markov chain on ${0,1}$ with $P(0\to1)=1$ and $P(1\to0)=1$.

Transition matrix:

$$ Q = \begin{pmatrix}0&1\1&0\end{pmatrix}. $$

This is strongly connected (you can reach any state from any state).

Initial distribution: $q^{(0)} = [1,0]^T$ (start at state 0 with certainty).

Then $q^{(t)} = Q^t q^{(0)}$:

$$ q^{(0)}=[1,0]^T, \quad q^{(1)}=[0,1]^T, \quad q^{(2)}=[1,0]^T, \quad \ldots $$

$q^{(t)}$ alternates between $[1,0]^T$ and $[0,1]^T$ forever, so $\lim_{t\to\infty} q^{(t)}$ does **not** exist.

Why? The chain is **periodic** with period 2. At even times all mass is at state 0; at odd times all mass is at state 1. The Fundamental Theorem requires the chain to be *aperiodic* (as well as irreducible) for $q^{(t)}$ to converge. While the long-run average $\bar q = \frac12[1,0]^T + \frac12[0,1]^T = [1/2,1/2]^T$ converges to the unique stationary distribution $\pi=[1/2,1/2]^T$, the sequence itself oscillates.

Problem 8 (5 pts) — HITS on Figure 2

Graph (from Figure 2): $A\to B$, $A\to C$, $A\to D$, $B\to A$, $B\to D$, $C\to A$, $D\to B$, $D\to C$.

Out-degrees: $A=3$, $B=2$, $C=1$, $D=2$. In-degrees: all equal 2.

Adjacency matrix L (rows = from, cols = to, order A,B,C,D):

$$ L = \begin{pmatrix} 0&1&1&1\ 1&0&0&1\ 1&0&0&0\ 0&1&1&0 \end{pmatrix}. $$

Authority scores — leading eigenvector of L^TL

$$ L^TL = \begin{pmatrix} 2&0&0&1\ 0&2&2&1\ 0&2&2&1\ 1&1&1&2 \end{pmatrix}. $$

Note: rows and columns B and C are identical, so $a_B = a_C$ in the leading eigenvector.

Antisymmetric eigenvector ($a_A=a_D=0$, $a_B=-a_C$): gives eigenvalue $\lambda=0$.

Symmetric subspace ($a_B=a_C=p$, unknowns a_A,p,a_D): the characteristic equation reduces to

$$ \lambda^3 - 8\lambda^2 + 17\lambda - 8 = 0, $$

with roots $\lambda \approx 4.814,\ 2.529,\ 0.657$ (no rational roots).

For the leading eigenvalue $\lambda_1 \approx 4.814$, the eigenvector satisfies $a_D=(\lambda_1-2)a_A$ and $p=(\lambda_1-2)a_A/(\lambda_1-4)$. Normalizing,

$$ a \approx \begin{pmatrix}0.1745\0.6035\0.6035\0.4910\end{pmatrix} \quad (A,B,C,D). $$

Hub scores — leading eigenvector of LL^T

$$ LL^T = \begin{pmatrix} 3&1&0&2\ 1&2&1&0\ 0&1&1&0\ 2&0&0&2 \end{pmatrix}. $$

Same nonzero eigenvalues as L^TL (both share $\lambda_1\approx 4.814$). The leading eigenvector is

$$ h \approx \begin{pmatrix}0.7739\0.3033\0.0795\0.5501\end{pmatrix} \quad (A,B,C,D). $$

Power iteration (first 3 rounds, unnormalized)

Update rule: $a \leftarrow L^T h$ then $h \leftarrow L a$ (normalize after convergence).

Start with

$$ a^{(0)} = h^{(0)} = \begin{pmatrix}1\1\1\1\end{pmatrix}. $$

Then the first three unnormalized rounds are

$$ a^{(1)} = \begin{pmatrix}2\2\2\2\end{pmatrix}, \qquad h^{(1)} = \begin{pmatrix}6\4\2\4\end{pmatrix}, $$

$$ a^{(2)} = \begin{pmatrix}6\10\10\10\end{pmatrix}, \qquad h^{(2)} = \begin{pmatrix}30\16\6\20\end{pmatrix}, $$

$$ a^{(3)} = \begin{pmatrix}22\50\50\46\end{pmatrix}, \qquad h^{(3)} = \begin{pmatrix}146\68\22\100\end{pmatrix}. $$

After iteration 3, the ratios $a_B:a_C:a_D:a_A = 50:50:46:22$ are converging toward the eigenvector $B=C > D \gg A$, and $h_A:h_D:h_B:h_C = 146 \overset{100}{=} 68:22$ shows $A \gg D > B \gg C$.

Interpretation:

- **A** has by far the highest hubbiness (out-degree 3, points to all others).

- **B and C** tie for highest authority: each is pointed to by 2 nodes (A and D), and D itself has high hubbiness.
- **D** has the second-highest hubbiness (out-degree 2, points to the high-authority nodes B,C) and second-highest authority (pointed to by A and B).
- **C** has the lowest hubbiness (out-degree 1, only points to A).

Problem 9 (15 pts) — PageRank on Figure 2

Same graph: $A\to B$, $A\to C$, $A\to D$, $B\to A$, $B\to D$, $C\to A$, $D\to B$, $D\to C$.

Every node has at least one outgoing edge, so there are **no dangling nodes**.

Transition matrix (row-stochastic):

$$ P = \begin{pmatrix} 0&\tfrac13&\tfrac13&\tfrac13\ \tfrac12&0&0&\tfrac12\ 1&0&0&0\ 0&\tfrac12&\tfrac12&0 \end{pmatrix}. $$

(a) PageRank without restart

Solve $\pi P = \pi$ (i.e. $\pi_j = \sum_i \pi_i P_{ij}$):

$$ \pi_A = \tfrac12\pi_B + \pi_C, \quad \pi_B = \tfrac13\pi_A + \tfrac12\pi_D, \quad \pi_C = \tfrac13\pi_A + \tfrac12\pi_D, \quad \pi_D = \tfrac13\pi_A + \tfrac12\pi_B. $$

From the equations for π_B and π_C, we get $\pi_B = \pi_C$. Then the symmetry between the equations for π_B and π_D gives $\pi_B = \pi_D$ as well.

Setting $b = \pi_B = \pi_C = \pi_D$, we get $\pi_A = \tfrac12 b + b = \tfrac32 b$. Normalizing,

$$ \tfrac32 b + b + b + b = \tfrac92 b = 1 \implies b = \tfrac29. $$

$$ \boxed{\pi_A = \frac{1}{3},\quad \pi_B = \pi_C = \pi_D = \frac{2}{9}.} $$

A has the highest PageRank, reflecting its role as the central hub with the highest out-degree. B, C, and D are equal by symmetry.

(b) PageRank with restart, $1-\beta = 0.20$ ($\beta = \tfrac45$, $n=4$)

$$ \pi_j = \tfrac45\sum_i \pi_i P_{ij} + \tfrac{1}{20}. $$

By the same symmetry argument as in part (a), let $b = \pi_B = \pi_C = \pi_D$. Then

$$ \pi_A = \tfrac45\left(\tfrac{b}{2} + b\right) + \tfrac{1}{20} = \tfrac65 b + \tfrac{1}{20}, $$

and

$$ b = \tfrac45\left(\tfrac{\pi_A}{3} + \tfrac{b}{2}\right) + \tfrac{1}{20} = \tfrac{4\pi_A}{15} + \tfrac{2b}{5} + \tfrac{1}{20}. $$

Substituting $\pi_A = \tfrac65 b + \tfrac{1}{20}$ gives

$$ b = \tfrac{4}{15}\left(\tfrac65 b + \tfrac{1}{20}\right) + \tfrac{2b}{5} + \tfrac{1}{20} = \tfrac{8b}{25} + \tfrac{1}{75} + \tfrac{2b}{5} + \tfrac{1}{20}. $$

Hence

$$ b\left(1 - \tfrac{8}{25} - \tfrac{2}{5}\right) = \tfrac{1}{75} + \tfrac{1}{20} = \tfrac{19}{300}, $$

so

$$ \tfrac{7b}{25} = \tfrac{19}{300} \implies b = \tfrac{19}{84}. $$

Therefore,

$$ \boxed{\pi_A = \frac{9}{28},\quad \pi_B = \pi_C = \pi_D = \frac{19}{84}.} $$

(c) Add 4 spam nodes v_1,v_2,v_3,v_4 with $v_i\to A$; restart $1-\beta=0.20$, $n=8$

Now $(1-\beta)/n = \tfrac{1}{40}$. Since no one links to any v_i,

$$ \pi_{v_i} = \tfrac45\cdot 0 + \tfrac{1}{40} = \tfrac{1}{40} \quad \text{for each } i. $$

Again let $b = \pi_B = \pi_C = \pi_D$. Then

$$ \pi_A = \tfrac45\left(\tfrac{b}{2}+b+4\cdot\tfrac{1}{40}\right)+\tfrac{1}{40} = \tfrac65 b + \tfrac{21}{200}, $$

and

$$ b = \tfrac{4\pi_A}{15}+\tfrac{2b}{5}+\tfrac{1}{40}. $$

Substituting for π_A gives

$$ \tfrac{7b}{25} = \tfrac{4}{15}\left(\tfrac65 b+\tfrac{21}{200}\right)-\tfrac{2b}{5}+\tfrac{1}{40} = \tfrac{7}{250}+\tfrac{1}{40} = \tfrac{53}{1000}. $$

Thus

$$ b = \tfrac{53}{1000}\cdot\tfrac{25}{7} = \tfrac{53}{280}, $$

and

$$ \pi_A = \tfrac65\cdot\tfrac{53}{280}+\tfrac{21}{200} = \tfrac{318}{1400}+\tfrac{147}{1400} = \tfrac{93}{280}. $$

So

$$ \boxed{\pi_A = \frac{93}{280},\quad \pi_B = \pi_C = \pi_D = \frac{53}{280},\quad \pi_{v_i} = \frac{1}{40} = \frac{7}{280}.} $$

Effect of link farm: A's PageRank increased from $\tfrac{9}{28}\approx 0.321$ to $\tfrac{93}{280}\approx 0.332$, only a modest gain. B,C,D dropped from $\tfrac{19}{84}\approx 0.226$ to $\tfrac{53}{280}\approx 0.189$ because the spam nodes absorb teleportation mass without contributing useful links to anyone but A.